



Exposure to pesticides and diabetes: A systematic review and meta-analysis



Evangelos Evangelou^{a,b}, Georgios Ntritsos^a, Maria Chondrogiorgi^a, Fotini K. Kavvoura^c, Antonio F. Hernández^d, Evangelia E. Ntzani^a, Ioanna Tzoulaki^{a,b,e,*}

^a Department of Hygiene and Epidemiology, University of Ioannina Medical School, Ioannina, Greece

^b Department of Biostatistics and Epidemiology, Imperial College London, London, UK

^c Oxford Centre for Diabetes, Endocrinology & Metabolism, University of Oxford, Churchill Hospital, Oxford, UK

^d University of Granada, School of Medicine, Granada, Spain

^e MRC-PHE Centre for Environment and Health, Imperial College London, UK

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ABSTRACT

Background: Diabetes mellitus has a multifactorial pathogenesis with a strong genetic component as well as many environmental and lifestyle influences. Emerging evidence suggests that environmental contaminants, including pesticides, might play an important role in the pathogenesis of diabetes.

Objectives: We performed a systematic review and meta-analysis of observational studies that assessed the association between exposure to pesticides and diabetes and we examined the presence of heterogeneity and biases across available studies.

Methods: A comprehensive literature search of peer-reviewed original research pertaining to pesticide exposure and diabetes, published until 30st May 2015, with no language restriction, was conducted. Eligible studies were those that investigated potential associations between pesticides and diabetes without restrictions on diabetes type. We included cohort studies, case-control studies and cross-sectional studies. We extracted information on study characteristics, type of pesticide assessed, exposure assessment, outcome definition, effect estimate and sample size.

Results: We identified 22 studies assessing the association between pesticides and diabetes. The summary OR for the association of top vs. bottom tertile of exposure to any type of pesticide and diabetes was 1.58 (95% CI: 1.32–1.90, $p = 1.21 \times 10^{-6}$), with large heterogeneity ($I^2 = 66.8\%$). Studies evaluating Type 2 diabetes in particular ($n = 13$ studies), showed a similar summary effect comparing top vs. bottom tertiles of exposure: 1.61 (95% CI 1.37–1.88, $p = 3.51 \times 10^{-9}$) with no heterogeneity ($I^2 = 0\%$). Analysis by type of pesticide yielded an increased risk of diabetes for DDE, heptachlor, HCB, DDT, and *trans*-nonachlor or chlordane.

Conclusions: The epidemiological evidence, supported by mechanistic studies, suggests an association between exposure to organochlorine pesticides and Type 2 diabetes.

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1. Introduction

Diabetes mellitus encompasses a group of diseases characterized by hyperglycaemia due to reduced insulin production, insulin action, or both. Type 2 diabetes (T2D) accounts for the vast majority of cases (approximately 90% of all cases) (Nolan et al., 2011). Diabetes is a world-wide epidemic, currently affecting more than 350 million people and expected to reach 550 million by 2050 (Whiting et al.,

2011). It represents a major public health challenge across the globe, due to the immediate cost of treatment but also the burden of diabetes-associated morbidity and mortality.

Diabetes mellitus has a multifactorial pathogenesis with a strong genetic component (Bradfield et al., 2011; Morris et al., 2012) as well as many lifestyle influences. The increasing incidence of diabetes, in particular T2D, has been mainly attributed to lifestyle factors, including westernization of diet and obesity. Over the last few years, emerging evidence suggests that environmental contaminants may also play an important role. Pesticides represent an increasingly widespread environmental exposure today and some of them (e.g. organochlorine [OC]) have the potential to accumulate in human tissues either through direct exposure or through the food chain. Different types of pesticides including OC compounds have been directly associated with increased

Abbreviations: OR, Odds ratio; CI, Confidence interval; DDE, Dichlorodiphenyl dichloroethylene; HCB, Hexachlorobenzene; DDT, Dichlorodiphenyltrichloroethane.

* Corresponding author at: Department of Biostatistics and Epidemiology, St Mary's Campus, Imperial College London, London, W2 1PG, UK.

E-mail address: i.tzoulaki@imperial.ac.uk (I. Tzoulaki).

T2D risk in a dose-response way (Lee et al., 2006) as well as with diabetes risk factors including adiposity, insulin resistance and dyslipidaemia (Lee et al., 2011b). However, evidence to date has been inconsistent.

In this report, we performed the first systematic review of observational studies that assess the association between exposure to a wide range of pesticides and different types of diabetes including T2D, Type 1 diabetes (T1D) and gestational diabetes. We performed quantitative synthesis to calculate a summary effect size when data were available and we assessed the presence of heterogeneity and biases across available studies.

2. Materials and methods

2.1. Data sources and searches

We conducted a comprehensive literature search of peer-reviewed original research pertaining to pesticide exposure and diabetes. We have updated a systematic literature search published by the European Food and Safety Authority, which only included studies published within January 2006 to September 2012, to include studies published before and after those dates (Ntzani et al., 2013), using the same search strategy and inclusion criteria. The review highlighted a potential association between diabetes and pesticides but concluded that evidence beyond the timeline examined needs to be gathered to support this observation along with more thorough examination of individual study characteristics and individual pesticides (Ntzani et al., 2013). The search strategy was designed so as to identify observational epidemiologic studies which examined the relationship between any pesticide exposure and any type of diabetes. The last search was performed on 30st May 2015. Two reviewers (GN, MC) independently searched MEDLINE (through PubMed) and EMBASE (See Fig. A.1 in Supplemental Material for search algorithm) without any language restrictions to identify eligible articles. In case of inconsistencies, consensus was reached with a third experienced reviewer (EE, IT, EEN). Studies were first screened at the title only level as previously suggested (Mateen et al., 2013) followed by abstract and title screening.

2.2. Study selection

Eligible studies were those that investigated potential associations between pesticide exposure and diabetes without restrictions on diabetes type. We included cohort studies, case-control studies, and cross-sectional studies and we excluded reviews, case reports, conference abstracts and ecological studies. Studies or analyses on arsenic, α , β , hexachlorocyclohexane (HCH), lead, dioxins (and dioxin-like compounds), polychlorinated biphenyls (PCBs), and polychlorinated dibenzofurans PCDFs) were excluded as they are not strictly classified as pesticides. Also, studies on poisoning and Agent Orange studies on very high exposure doses were excluded.

2.3. Data extraction

From each eligible study, two independent reviewers extracted information on the first author; the journal and the year of publication; the type and name of pesticide assessed; how exposure was assessed (questionnaire/biomarker); the outcome definition; the effect estimate and its uncertainty; the comparison level; origin of the population; total sample size and number of cases and controls; and the study design.

2.4. Data synthesis and analysis

We defined three groups of studies based on their definition of the outcome examined: studies with no clear reported definition of diabetes type (not specific statement about type of diabetes) which according to the age range of the population could be assumed to be predominately T2D studies (nonspecific definition); and studies with a clear

statement in their methods that they studied T2D type (specific definition); studies of T1D; and studies of gestational diabetes. Our main analyses focused on the association of T2D (either nonspecific or specific) with the exposure to i) any pesticide and ii) OC pesticides only. We also performed a sensitivity analysis including only the studies that provide estimates adjusted for BMI and/or smoking habits. Eligible studies often reported more than one analysis using different types of pesticides in the same article. For the analysis on any pesticide, we included in the main analysis the reported comparison with the largest sample size opting for maximal precision. Whenever two or more eligible articles studied the same population, pesticide and examined the same exposure period, the most recent publication was included in the analysis. Additionally, we performed meta-analyses for specific types of pesticides whenever data were available in more than 3 studies. We calculated summary ORs by pooling the study-specific estimates using fixed and random-effects models (DerSimonian and Laird, 1986; Lau et al., 1997). The presence and extent of heterogeneity was assessed by the I^2 (ranging from 0% to 100%) (Higgins et al., 2003; Ioannidis et al., 2007). We assessed 95% predictive intervals (PI) to display the confidence interval of the approximate predictive distribution of a future trial, based on the extent of heterogeneity (Higgins et al., 2009; Riley et al., 2011).

Different effect estimates were often used for the association between pesticides exposure and diabetes across studies (such as per-unit change or per-1-SD change or comparisons of the extremes of quintiles, quartiles, tertiles or other groupings). To enable a consistent and comparable approach to meta-analysis, estimates were transformed and harmonized to represent top and bottom tertiles using previously described and widely used methods. Harmonization to unit increase changes was not possible as most studies reported categorical and not continuous exposure estimates. For example, log risk estimates were transformed with the comparison between the top and the bottom thirds being equal to 2.18 times the log risk estimate for one SD increase, 2.54 times for a comparison of the top vs. bottom quartiles and 2.78 for the top vs. bottom quintiles (Chene and Thompson, 1996).

We assessed small study effects (an indication of publication bias) by visual inspection of funnel plots and the Egger test (Egger et al., 1997). To assess possible sources of heterogeneity we performed a meta-regression according to pre-specified study-level characteristics: exposure assessment; exposure type (occupational vs. non-occupational); and median sample sizes (above vs. below). All analyses were performed with Stata (version 10; StataCorp, College Station, TX, USA) and p -values < 0.05 were deemed significant.

3. Results

Fig. 1 shows the flow chart for the study selection process; nine new studies updated the systematic review prepared for and published by EFSA (Ntzani et al., 2013). In total, 25 studies assessing diabetes were deemed eligible with a total sample size of 80,161 participants (5841 cases/74,320 controls) and median sample size of 725 participants (interquartile range [IQR]: 352 to 2047) per study. Table 1 provides an overview of the main study characteristics. The majority of the studies were cross-sectional ($N = 12$ studies) but there was also data from 8 prospective cohort studies. Ten studies were based in North-America, 8 in Europe, 4 in Asia-Pacific region. The majority ($N = 21$) of studies assessed organochlorine exposure using biomarker measurements of various metabolites. Overall, 22 studies were either assumed to examined T2D due to study age range or clearly defined in their methods that examined T2D only, 1 study examined T1D only and 2 focused on gestational diabetes. All studies reported age-adjusted effects.

3.1. Type 2 diabetes

The summary OR for the association of the top vs. bottom tertile of exposure to any type of pesticide and T2D (nonspecific definition)

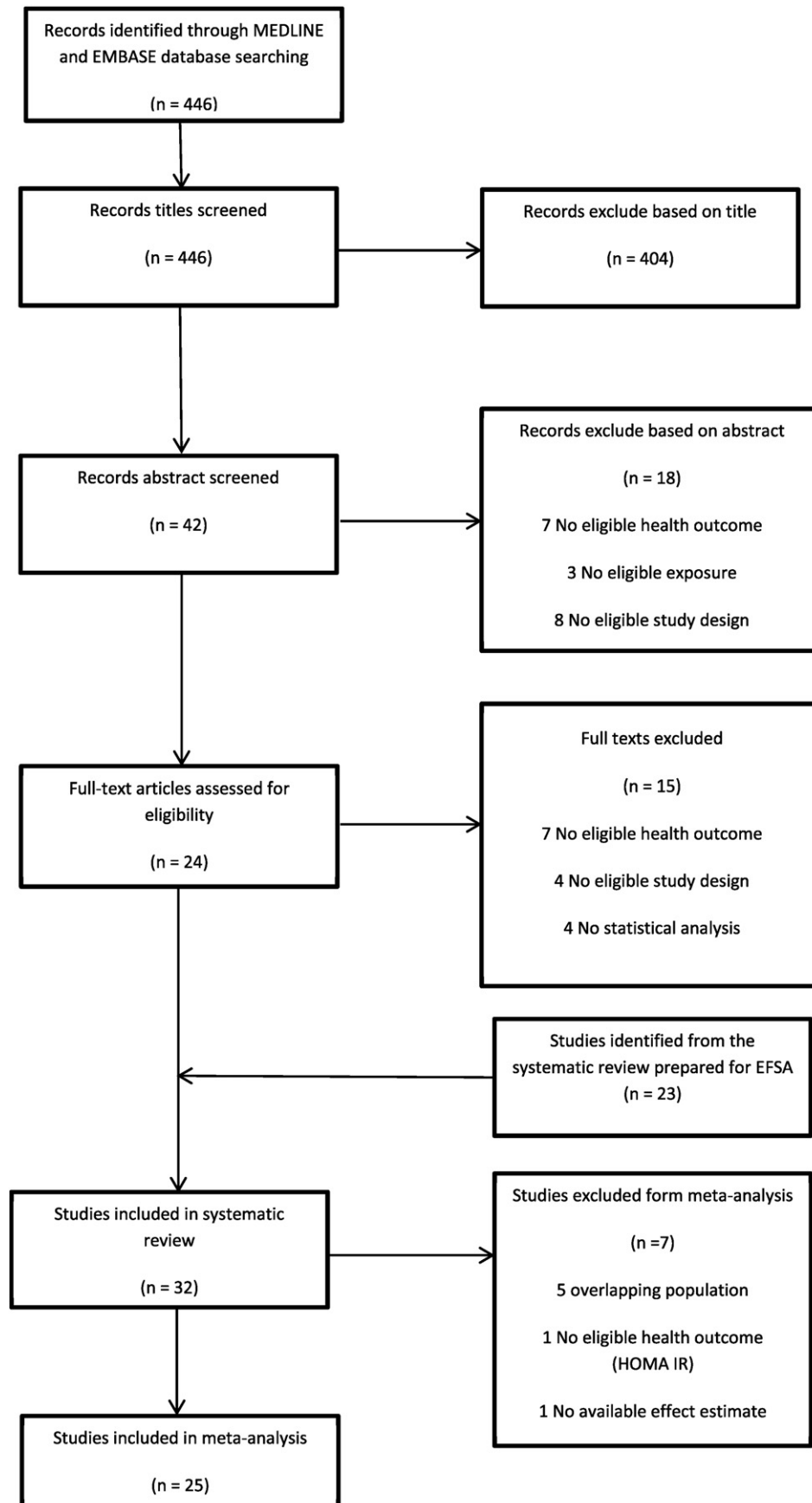


Fig. 1. Flow chart of the study selection process.

Table 1

Main characteristics of studies evaluating the association between pesticides exposure and diabetes.

Study, year (Reference)	Sample size	Study design	Region	Type of exposure	Diabetes definition ^a	Source of pesticide assessment	Pesticides assessed
Rylander et al. (2005)	253	Cross-sectional	Sweden	Environmental	T2D	Biomarker levels in serum	p,p'-DDE
Codru et al. (2007)	352	Cross-sectional	USA	Both	T2D (non specific definition)	Biomarker levels in serum	DDE, HCB, mirex
Cox et al. (2007)	1303	Cross-sectional	USA	Occupational	T2D (non specific definition)	Biomarker levels in serum	p,p'-DDT, p,p'-DDE, dieldrin, oxychlordane, β -hexachlorocyclohexane, hexachlorobenzene, and <i>trans</i> -nonachlor
Rignell-Hydbom et al. (2007)	272	Cross-sectional	Sweden	Environmental	T2D	Biomarker levels in serum	p,p'-DDE
Jorgensen et al. (2008)	692	Cross-sectional	Greenland	Environmental	T2D	Biomarker levels in serum	Organochlorine insecticides
Philibert et al. (2009)	101	Cross-sectional	Canada	Environmental	T2D (non specific definition)	Biomarker levels in serum	p,p'-DDE
Ukropec et al. (2010)	819	Cross-sectional	Slovakia	Environmental	T2D (non specific definition)	Biomarker levels in serum	HCB, p,p'-DDE, p,p'-DDT
Patel et al. (2010)	2545	Cross-sectional	USA	Environmental	T2D	Biomarker levels in serum	Heptachlor
Airaksinen et al. (2011)	398	Cross-sectional	Finland	Environmental	T2D	Biomarker levels in serum	Oxychlordane, <i>trans</i> -nonachlor, p,p'-DDE
Arrebola et al. (2013)	386	Cross-sectional	Spain	Environmental	T2D	Biomarker levels in serum	HCB, p,p'-DDE, SUM OCs
Kim et al. (2013)	33	Cross-sectional	Korea	Environmental	T2D	Biomarker levels in adipose tissue	DDTs, chlordane, HCB
Al-Othman et al. (2015)	280	Cross-sectional	Saudi Arabia	Both	T2D	Biomarker levels in serum	2,4-DDE, 4,4-DDE, 4,4-DDD, 2,4-DDT, 4,4-DDT, sum DDTs
Rignell-Hydbom et al. (2009)	742	Case-control	Sweden	Environmental	T2D	Biomarker levels in serum	p,p'-DDE
Son et al. (2010)	62	Case-control	Korea	Environmental	T2D	Biomarker levels in serum	Oxychlordane, <i>trans</i> -nonachlor, heptachlor epoxide, hexachlorobenzene, beta--hexachlorocyclohexane, mirex, p,p'-DDE, p,p'-DDD, p,p'-DDT, o,p'-DDT
Lee et al. (2010)	103	Nested case-control	USA	Environmental	T2D	Biomarker levels in serum	Oxychlordane, <i>trans</i> -nonachlor, hexachlorobenzene, beta--hexachlorocyclohexane, gamma--hexachlorocyclohexane, mirex, p,p'-DDE, p,p'-DDT
Morgan et al. (1980)	2597	Cohort	USA	Occupational	T2D (non specific definition)	Biomarker levels in serum	Organochlorine pesticides
Beard et al. (2003)	44	Cohort	Australia	Occupational	T2D (non specific definition)	Questionnaire	DDT
Montgomery et al. (2008)	15,430	Cohort	USA	Occupational	T2D (non specific definition)	Questionnaire	2,4-dichlorophenoxyacetic acid (2,4-D), glyphosate, fonofos, diazinon, imazethapyr, butylate, petroleum oil, trichlorfon, DDT, parathion, phorate, terbufos, alachlor, pendimethalin, chlorpyrifos, benomyl, heptachlor, aldicarb, carbon tetrachloride, permethrin, dieldrin, cyanazine, metalaxyl, aluminum phosphide, dicamba, aldrin, EPTC, maneb, malathion, 2,4,5-T, 2,4,5-TP, atrazine, ethylene dibromide, metolachlor, permethrin, dichlorvos, carborufan, carbamate, captan, chlordane, metribuzin, trifluralin, coumaphos, chlorothalonil, toxaphene, chlorimuron ethyl, paraquat, methyl bromide, ziram, carbaryl
Turyk et al. (2009)	309	Cohort	USA	Environmental	T2D (non specific definition)	Biomarker levels in serum	DDE
Lee et al. (2011a)	287	Cohort	Sweden	Environmental	T2D	Biomarker levels in serum	p,p'-DDE, <i>trans</i> -nonachlor, hexachlorobenzene, OC
Starling et al. (2014)	13,637	Cohort	USA	Occupational	T2D (non specific definition)	Questionnaire	Aldrin, chlordane, DDT, dieldrin, heptachlor, lindane, toxaphene, chlorpyrifos, coumaphos, diazinon, dichlorvos, fonofos, malathion, parathion, phorate, terbufos, aldicarb, carbaryl, carborufan, permethrin, 2,4,5-T, 2,4,5-TP, 2,4-D, alachlor, atrazine, butylate, chlorimuron-ethyl, cyanazine, dicamba, EPTC, glyphosate, imazethapyr, metolachlor, metribuzin, paraquat, pendimethalin, petroleum oil, trifluralin, benomyl, captan, chlorothalonil, maneb, metalaxyl, methyl bromide, carbon tetrachloride/carbon disulfide
Wu et al. (2013)	729	Case-control	USA	Environmental	T2D	Biomarker levels in serum	Hexachlorobenzene, p,p'-DDE, p,p'-DDT
Rignell-Hydbom et al. (2010)	159	Nested case-control	Sweden	Environmental	T1D	Biomarker levels in serum	p,p'-DDE

(continued on next page)

Table 1 (continued)

Study, year (Reference)	Sample size	Study design	Region	Type of exposure	Diabetes definition ^a	Source of pesticide assessment	Pesticides assessed
Saunders et al. (2014)	779	Cohort	Gouadeloupe	Environmental	Gestational diabetes	Biomarker levels in serum	Chlordecone
Saldana et al. (2007)	7055	Cohort	USA	Environmental	Gestational diabetes	Questionnaire	2,4-D, 2,4,5-TP, 2,4,5-T, alachlor, atrazine, butylate, cyanazine, dicamba, glyphosate, pendimethalin, petroleum oil, trifluralin, diazinon, malathion, terbufos, phorate, carbaryl, carborufan

^a These studies were assumed to study T2D only (and not T1D) as due to the age range of the population the overwhelming majority of cases of diabetes would be T2D. The rest of the studies classified as T2D clearly mentioned in their methods that the diabetes cases were T2D only.

was 1.58 (95% CI: 1.32–1.90, p -value = 1.2×10^{-6}) using random effects models (Table 2; Fig. 2); for exposure to OC pesticides the summary OR was 1.68 (95% CI: 1.37–2.07). Large heterogeneity was observed: $I^2 = 66.8\%$ and 69.7% respectively. The 95% PI were 0.86–2.96 and 0.81–3.49, including the null in both cases. Studies which clearly defined in their methods that they study T2D only showed similar results; All these studies referred to OC pesticides only and the summary OR comparing top vs. bottom tertile of exposure was 1.61 (95% CI: 1.37–1.88, p -value = 3.5×10^{-9}) and no heterogeneity was observed ($I^2 = 0\%$) (Table 2). The 95% PI was 1.35–1.92, excluding the null. The summary effect estimates for BMI adjusted studies were 1.61 ($n = 16$ studies, 95% CI: 1.29–2.00, p -value = 2.1×10^{-5}) and 1.59 ($n = 10$ studies, 95% CI: 1.33–1.90, p -value = 4.1×10^{-7}) for the nonspecific and the specific definition respectively. For smoking adjusted estimates the relevant summary effects were 2.36 ($n = 10$ studies, 95% CI: 1.46–4.82, p -value 4.8×10^{-4}) and 1.78 ($n = 6$ studies, 95% CI: 1.13–2.81, p -value = 0.013).

Analysis by type of pesticide yielded an increased risk of diabetes for DDE, heptachlor, HCB, DDT, and *trans*-nonachlor or chlordane. Overall, less heterogeneity was observed in these pesticide-specific analyses which was further reduced when analyses were limited to studies with clear/specific diabetes type definition (Table 2). Specifically, the smallest heterogeneity was observed for HCB ($I^2 = 22\%$, $n = 7$ studies, OR = 1.79, 95% CI = 1.26–2.54, top vs. bottom tertile) whereas the largest heterogeneity was assessed for exposure to DDT

($n = 9$ studies) and dieldrin ($n = 3$ studies) where the heterogeneity was 80.1% and 82.4% respectively. In all other cases heterogeneity was small to moderate. The summary effects ranged from 1.47 (95% CI: 1.15–1.86) for top vs. bottom tertile of exposure to chlordane and oxychlordane and *trans*-nonachlor ($n = 8$ studies), up to 1.95 (95% CI: 1.44–2.66) for exposure to DDE ($n = 14$ studies). The strongest association was shown for DDE with a summary OR of 1.95 (95% CI: 1.44–2.66) and p -value = 1.7×10^{-5} comparing top and bottom tertiles of exposure.

3.2. Small study effect

Egger's test was statistically significant ($p = 0.003$) suggesting a prominent small study effect. A sensitivity analysis using the median sample size as a cut-off for the eligible comparisons showed that the effect was indeed larger in the smaller studies group; however, their difference was not significant, (summary OR top vs. bottom tertile 2.44 (95% CI: 1.60–3.73, p -value = 0.0003 vs. 1.38 (95% CI: 1.15–1.67, p -value = 0.001). For the specific definition of the outcome, Egger's test was not statistically significant ($p = 0.269$) and again, the smaller studies had inflated but not significantly different effect sizes compared to larger studies (summary OR top vs. bottom tertile 1.93, 95% CI: 1.33–2.81, p -value = 0.001 vs 1.55, 95% CI: 1.30–1.84, p -value = 9×10^{-7}) (Table 3). Further subgroup analyses according to study design, measurement of the exposure, type of exposure, median of study population

Table 2
Summary odds ratios (OR) and 95% CI by random effects meta-analysis for different pesticide comparisons stratified by the exposure of the T2D: overall (non specific) and specific (clearly defined T2D).

	N studies	Summary OR (95% CI)	P value	I ² %
Any pesticide				
T2D (non specific definition) ^a	22	1.58 (1.32, 1.90)	1.2×10^{-6}	67
T2D (specific definition)	13	1.61 (1.37, 1.88)	3.5×10^{-9}	0
Any OC pesticide				
T2D (non specific definition) ^a	22	1.68 (1.37, 2.07)	8.1×10^{-7}	69.7
T2D (specific definition)	13	1.61 (1.37, 1.88)	3.5×10^{-9}	0
Chlordane and oxychlordane and <i>trans</i> -nonachlor				
T2D (non specific definition) ^a	8	1.47 (1.15, 1.86)	0.002	45
T2D (specific definition)	5	1.79 (1.31, 2.43)	2.3×10^{-4}	0
DDT and p,p'-DDT				
T2D (non specific definition) ^a	9	1.79 (1.28, 2.49)	0.001	80
T2D (specific definition)	5	2.06 (1.05, 4.04)	0.036	53
DDE and p,p'-DDE				
T2D (non specific definition) ^a	14	1.95 (1.44, 2.66)	1.7×10^{-5}	45
T2D (specific definition)	9	1.65 (1.15, 2.37)	0.006	39.5
Dieldrin				
T2D (non specific definition) ^a	3	1.78 (0.80, 3.96)	0.159	82
T2D (specific definition)	0			
HCB				
T2D (non specific definition) ^a	7	1.79 (1.26, 2.54)	0.001	22
T2D (specific definition)	4	2.21 (1.27, 2.83)	0.005	0
Heptachlor				
T2D (non specific definition) ^a	4	1.57 (1.16, 2.13)	0.003	66
T2D (specific definition)	2	1.73 (1.37, 2.19)	5.2×10^{-6}	0

^a These studies were assumed to study T2D only (and not T1D) as due to the age range of the population the overwhelming majority of cases of diabetes would be T2D. The rest of the studies classified as T2D (specific definition) clearly mentioned in their methods that the diabetes cases were T2D only.

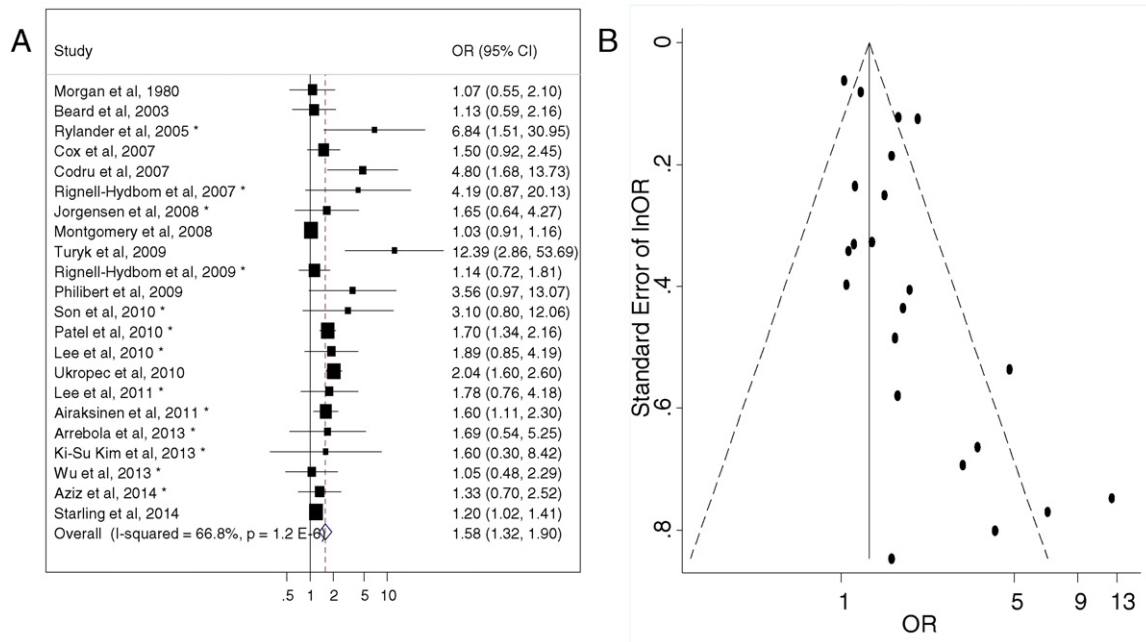


Fig. 2. Forest plot showing the standardized effect sizes comparing top vs. bottom tertiles of exposure across all eligible studies on studies with non specific and specific (*) definition of T2D (A) and respective funnel plot (B).

and continent where the study was performed did not reveal any differences in the summary effect sizes (Table 3).

3.3. Other types of diabetes

Our systematic review revealed one additional study that assessed only T1D accumulating a total sample size of 300 participants (150 cases and 150 controls), showing a non-significant association (OR = 0.71, 95% CI 0.37–1.34 for top vs bottom tertile of exposure).

Also, we identified two additional studies that assessed gestational diabetes and exposure to pesticides. Both studies showed non-significant associations. Specifically, a large prospective cohort study with 7055 participants including 317 women with gestational diabetes who were

exposed to pesticides compared to those not exposed shown an OR = 1.0 (95% CI 0.8–1.3) (Saldana et al., 2007). Additionally, a cohort study with 779 participants of which 71 were cases diagnosed with gestational diabetes had an OR = 0.74 (95% CI 0.37–1.48, $N = 779$) comparing top and bottom quartiles of exposure (Saunders et al., 2014).

4. Discussion

To our knowledge this is the largest comprehensive systematic review and meta-analysis including all types of pesticide exposure and all types of diabetes. The findings of our systematic review support the hypothesis that exposure to organochlorine (OC) pesticides increases the risk of T2D. In fact, exposure to different types of OC pesticides, including chlordane,

Table 3
Summary odds ratios (OR) and 95% CI by random effects meta-analysis stratified by different study characteristics.

	N	Summary OR (95% CI)	P value	I ² %		N	Summary OR (95% CI)	P value	I ² %
	T2D non specific definition					T2D specific definition			
<i>Study design</i>									
Cross sectional	12	1.83 (1.58, 2.12)	1.8×10^{-15}	5	8	1.69 (1.41, 2.03)	1.3×10^{-8}	0	
Case-control	4	11.32 (0.93, 1.86)	0.120	0.6	4	1.32 (0.93, 1.86)	0.120	0.6	
Prospective	6	1.22 (0.95, 1.55)	0.114	64	1	1.78 (0.76, 4.18)	0.185	NP	
<i>Pesticide assessment</i>									
Biomarker	19	1.75 (1.46, 2.10)	1.7×10^{-9}	32	13	1.61 (1.37, 1.88)	3.5×10^{-9}	0	
Questionnaire	3	1.10 (0.98, 3.22)	0.097	12		NA			
<i>Type of exposure</i>									
Environmental	15	1.81 (1.48, 2.20)	5.2×10^{-9}	29	12	1.63 (1.38, 1.91)	4.3×10^{-9}	0	
Occupational or both	5	1.10 (1.01, 1.21)	0.038	0	1	1.33 (0.70, 2.52)	0.383		
<i>Study sample size</i>									
Sample size ≤ 369	11	2.44 (1.60, 3.73)	0.0003	44	7	1.93 (1.33, 2.81)	0.001	0	
Sample size > 369	11	1.38 (1.15, 1.67)	0.001	72	6	1.55 (1.30, 1.84)	9×10^{-7}	0	
<i>Geographical area</i>									
America	10	1.50 (1.17, 1.92)	0.001	75	3	1.65 (1.32, 2.06)	8.5×10^{-6}	0	
Europe	8	1.76 (1.38, 2.24)	5.1×10^{-6}	25	7	1.61 (1.20, 2.17)	0.002	16	
Asia-Oceania	4	1.36 (0.90, 2.27)	0.147	0	3	1.56 (0.90, 2.69)	0.113	0	

NP: Not pertinent.

oxylchlordane and *trans*-nonachlor, heptachlor, DDT, DDE, dieldrin, HCB, were all associated with significantly higher risk of T2D.

The summary risk estimates for different classes of OC pesticides corresponded to relatively large effects and ranged from 1.47 (chlordane, oxylchlordane and *trans*-nonachlor) to 1.95 (DDE) comparing top vs bottom tertiles of exposure. However, those estimates need to be interpreted with caution given the inherent limitations of epidemiological data. In particular, most of the evidence results from case-control and cross-sectional studies where a temporal association cannot be shown and the possibility of reverse causality cannot be excluded. Nonetheless, summary of cohort studies also resulted in statistically significant effects sizes. Most importantly, in our study, we found evidence for small study effects with small studies with negative results largely missing. However, again subgroup analysis of large studies only showed more conservative results which were still statistically significant suggesting an increased risk of T2D with exposure to OC pesticides. Finally, a large amount of the accumulated evidence came from studies where the type of diabetes is not clearly defined; however, given the age range of the participants it is estimated that around 90% of the cases in those studies are indeed T2D (Nolan et al., 2011). Analyses using a narrow definition and limited to T2D cases only, still provided significant results with no heterogeneity observed despite the smaller amount of evidence. Finally, studies on T1D and gestational diabetes did not show any effect, however only a limited number of studies were included and therefore their results should be interpreted with caution.

Humans show background exposure (long-term low-dose) throughout the lifetime to OC compounds, including chlorinated pesticides. As these chemicals are lipophilic, hydrophobic and highly resistant to metabolic degradation, they are bioaccumulated in fatty tissues for many years, and their serum concentration is considered to be a good reflection of lifetime exposures (Lee et al., 2014). As the production and use of most OC pesticides were banned in Western countries four decades ago, their average absolute levels are decreasing in most of the population compared to earlier years. Nevertheless, OC pesticides are continuously released from fat stores to the blood and reach critical organs where can disturb some biological functions (Lee et al., 2014; Ruzzin et al., 2010).

The biological mechanism underlying the linkage of pesticide exposure to the pathogenesis of T2D remains largely unknown. The available epidemiological and experimental evidence suggests that background contamination from OC pesticides is strongly associated with the development of T2D (Lee et al., 2014). OC pesticides have variable molecular and cellular targets and thus they cannot be considered to have a single mode of action. Inflammation in adipose tissue, ectopic lipid accumulation (lipotoxicity) in liver, muscle and pancreas, and mitochondrial dysfunction are primary mechanisms underlying the pathogenesis of T2D and all of them have been associated with OC pesticides (Lee et al., 2014).

Experimental studies in rats fed salmon oil containing environmental levels of persistent organic pollutants, including OC pesticides, resulted in insulin resistance, visceral obesity, dyslipidemia, non-alcoholic fatty liver and chronic low-grade inflammation (Ruzzin et al., 2010). Further *in vitro* studies showed a reduced insulin action after treatment with OC pesticides, with this effect being observed at low doses but not at high doses, suggesting non-monotonic dose-response curve rather than a clear dose-response relation (Ruzzin et al., 2010). OC pesticides, like other persistent organic pollutants, may affect pancreatic β cells and trigger insulin resistance, thus impairing both lipid and glucose metabolism. The powerful inhibitory effect of these compounds on insulin action likely explains the common line of evidence from several independent studies reporting an association between T2D and the body burdens of p,p'-DDE, oxylchlordane, or *trans*-nonachlor. These findings support the notion that OC pesticides are potential contributors to the rise in prevalence of insulin resistance which subsequently partly explains the increase in T2D, in particular (Ruzzin

et al., 2010). Potential mechanisms underlying the relationship between OC pesticides exposure and T2D include binding to peroxisome proliferator activated receptors (PPAR) (Casals-Casas and Desvergne, 2011; Lau et al., 2010) or other nuclear receptors involved in the regulation of glucose and lipid metabolism and insulin sensitivity such as estrogen receptors (ERs) (Mauvais-Jarvis et al., 2013; Alonso-Magdalena et al., 2008) or the pregnane X receptor (PXR) and the constitutive androstane receptor (CAR) (Gao and Xie, 2012) (Timsit and Negishi, 2007). An alternative or additional mechanism to account for the potential of OC pesticides to cause T2D is oxidative damage. Leading to impairment of mitochondrial function and development of insulin resistance and T2D (Lee et al., 2014).

Our study has some limitations. The differences in the design of the studies, the type of the exposure and its measurements may be depicted in the effect sizes of pesticides in diabetes and can introduce heterogeneity in the meta-analysis. However, subgroup analyses did not reveal any significant differences in the risk estimates based on the type of studies, and the type or the measurement of the exposure. A small study effect bias may exist, leading to an inflated summary effect when we pooled all studies together. This is an indication that true effect may be closer to the effect provided from the larger studies in the field as sensitivity analysis shown. Indeed, the summary effect of the larger studies (based on the median sample size) was almost 43.5% smaller compared to all studies together (1.38 vs 2.44) with the non-specific definition and almost 19.7% smaller when the specific definition was used (1.55 vs 1.93). Other limitations include possible overestimation of associations due to preferential publication of extreme findings, selective reporting of results with striking associations, especially when multiple pesticides are considered. To overcome these caveats we recorded any available information from each study and we also standardized the comparison arms to avoid the synthesis of the extremes only. Another limitation pertains to the fact that only linear effects have been considered and the presence of potential non-linear effects may affect significantly the summary results. In most of the studies, some participants may not be totally unexposed which may underestimate the calculation of the true effect. Finally, we should take into consideration that high correlations among serum concentrations of various OC pesticides and other lipophilic chemicals and pollutants may exist and residual confounding cannot be ruled out. In this particular case, the measurement and the assessment of the effect sizes should be interpreted with caution as these chemicals may affect the magnitude of the observed association.

5. Conclusion

This systematic review provides a comprehensive synthesis of available evidence by including data from different sources and quantifying the risk for diabetes for a wide range of pesticides. Generalizability was achieved by including more than 300,000 participants from over 10 countries and 3 continents. The goal of this systematic research was to evaluate all types of pesticides examined in relation to diabetes; however, our exhaustive search strategy revealed that mostly OC pesticides have been studied to date. Evidence on several other types of pesticides is still limited and should be further considered in future studies. The findings from this analysis, besides any potential biases derived from the synthesis of observational studies that may inflate the summary risk estimate yielded supportive evidence that exposure of pesticides increase the risk of T2D. Larger, prospective studies may show more conservative risk effects; however, it seems that the role of pesticides is still harmful. This epidemiological evidence is supported by mechanistic studies, which suggest potential mechanisms through which these compounds may affect diabetes risk. The pathogenetic mechanisms under this association need to be further assessed experimentally and guidelines on use of pesticides and risk of diabetes should be considered in order to reflect current evidence.

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Conflict of interest statement

None declared.

Contribution statement

IT, EE and EEN have conceived the study. GN and MC have performed the literature search and statistical analyses. IT, AJH and EE have written the manuscript. All authors have made critical revisions to the manuscript.

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